

Applied A.I. Solutions

Foundations of Data Management

Professor
Daniel Vitaver, B.Sc., EMBA

daniel.vitaver@georgebrown.ca

COURSE OVERVIEW

COURSE DESCRIPTION

This course will provide an introduction to data management principles. The focus of this course will be on the policies, standards, and processes of a data governance framework. Students will apply data management best practices in areas such as metadata, master data and data quality management, root cause analysis, process design, data modeling and design, data integrity, and data privacy.

COURSE OUTCOMES

- 1. Explain the Data Management process in a data-driven organization (DAMA).**
2. Build a data governance framework for an organization.
3. Design a Data Architecture for efficient storage and operations.
4. Model and design the data required for a use-case.
5. Build data quality measures.

Detailed Evaluation System

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	Final Grade:
Participation	Attendance and in-class participation in various activities	1, 2, 3, 4, 5	1, 2, 4, 5, 6, 7, 8, 9, 10, 11	1 - 3	10
Lab Exercises 3 @ 10%	Group Assignment Hands-on exercises	1, 2, 3, 4, 5		1 - 3	30
Lab quizzes 5 @ 2%	Questions to evaluate students' pace on keeping up with the class			1-3	10
Project	Group Assignment			3	30
Presentation and/or Questionnaire	Project presentation and/or Q&A to evaluate the overall knowledge of the students	1, 2, 3, 4, 5	1,2,4,5,6,7,8,9, 10,11	3	20
TOTAL					100%

GRADING SYSTEM the passing grade for this course is: D (50%)

A+	90-100	4.0	B+	77-79	3.3	C+	67-69	2.3	D+	57-59	1.3	Below 50	F	0.0
A	86-89	4.0	B	73-76	3.0	C	63-66	2.0	D	50-56	1.0			
A-	80-85	3.7	B-	70-72	2.7	C-	60-62	1.7						

TOPICAL OUTLINE

Week	Topic Task	In-person / Online	Outcomes	Content / Activities	Resources
1	1	Monday Online	1, 2, 3	- Introduction to Data Management	Resource material available on D2L Brightspace
1	2	Wednesday Online	1, 2, 3	- Data Governance	Resource material available on D2L Brightspace
1	3	Friday In-person	1, 2, 3	- Reference and Master Data Management - Metadata Management	Resource material available on D2L Brightspace
					Week 1 Tasks: Lab Exercise 1 Lab Quiz 1, 2

TOPICAL OUTLINE

Week	Topic Task	In-person / Online	Outcomes	Content / Activities	Resources
2	4	Monday In-person	1, 2, 3, 4	- Data Architecture - Data Modeling and Design	Resource material available on D2L Brightspace
2	5	Wednesday Online	1, 2, 3, 4	- Data Integration and Interoperability	Resource material available on D2L Brightspace
2	6	Friday In-person	1, 2, 3, 4	- Data Storage and Operations	Resource material available on D2L Brightspace
		Week 2 Tasks: Lab Exercise 1+2 Lab Quiz 3, 4, 5			

TOPICAL OUTLINE

Week	Topic Task	In-person / Online	Outcomes	Content / Activities	Resources
3	7	Monday In-person	1, 2, 3, 4, 5	- Data Quality - Data Security	Resource material available on D2L Brightspace
3	8	Wednesday Online	1, 2, 3, 4, 5	- Document and Content Management - Data Privacy	Resource material available on D2L Brightspace
3	9	Friday In-person	1, 2, 3, 4, 5	- Project Due - Final Questionnaire	Resource material available on D2L Brightspace
		Week 3 Tasks: Lab Exercise 3 Project Due Final Questionnaire			

Excerpt from the College Policy on Academic Dishonesty:

The minimal consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material. To view George Brown College policies please go to www.georgebrown.ca/policies

*The “Foundations of Data Management” course can lead to a **CDMP** (Certified Data Management Professional) certification by DAMA*

Reference Book - Most recommended reading

- **DAMA DMBOK v2** - DAMA Data Management Knowledge System Guide (Original Book 2nd Edition or newest) - DAMA International – DMBOK v2 - Technics Publications Basking Ridge, New Jersey, USA
- <https://www.dama.org/cpages/dama-recommended-books>

